

Tutorial Letter 102/0/2026

Introduction to Programming II

COS1512

Assessment 2

Year Module(s): 2026

Name of Department: Computer Science

IMPORTANT INFORMATION

Please register on myUnisa, activate your myLife e-mail account and make sure that you have regular access to the myUnisa module website, MODULE COS1512-2026-Y, as well as your group website.

Note: This is a fully online module. It is, therefore, only available on myUnisa.

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1. Introduction

Kindly note the total mark for this assessment is [85], i.e. **Total = 85**. This assessment covers topics studied from chapters 4, 5, 6, 7, 10, and 14 of the prescribed books. The assessment consists of four questions.

2. Assignment 2 Instructions

Assignment Instructions: Workshop Peer- Assessment

Assignment 2 is a **four-phase assessment**, and you must note and act according to the requirements for each phase. The instructions for each phase are as follows:

Phase 1 – The File Submission Phase

- **Complete the questions.** Write the code to answer the question.
- While developing the solution, use comments on the code to indicate the purpose of the functions implemented.
- Answer questions in sequence or given order. The solution should be provided or done in a Word document and converted to PDF for submission. Copy and paste the code to the document including the screenshot of the output. **Both the code and comments and screenshot of the output must be submitted in a PDF file.**
- You must save the PDF document with your student number and the assessment as the name e.g. 12345678_A2.pdf.

File Submission

- The assessment must be uploaded as a single PDF document.

The grading phase will open as soon as the submission phase has closed, you will have one **(1) week to complete the grading phase.**

Phase 2 - Testing

Once phase 1 (submission) is closed, you are required to return to the assessment.

Download at least three (3) of your fellow student's submissions, and follow the instructions provided to see if you can solve the programming problem.

The testing/grading phase will open as soon as the submission phase has closed, **you will have one (1) week to complete the testing/grading phase.**

Phase 3 - Grading

Return to the assessment for **phase 3 and grade the three (3) students' step-by-step instructions, as well as your own, using the rubric that will be provided.** You will thus **grade four (4) assessments in total.**

Your result will be dependent on how you grade yourself and your fellow students, as well as how you are graded. If you grade too high or too low in comparison with the other students who grade the same assessment, your grading will negatively impact your own grade. You thus need to make sure you provide a fair and honest grade.

The testing/grading phase will open as soon as the submission phase has closed, you will have one (1) week to complete the testing/grading phase.

Phase 4

Once the assessment phase is closed, return to your assessment. **Read the feedback and comments provided by your fellow students.** If you have any queries related to your feedback, please send an email to the lecturer detailing why you do not agree with the assessment you received. **You have one (1) week after the close of the grading phase to request moderation of your grade.**

3. Questions

Question 1 [12]

- 1.1. The number used to refer to a particular element of an array is called its _____. (1)
- 1.2. State whether the following are *true* or *false*. If the answer is *false*, explain why.
- a) An array can store many different types of values. (1)
 - b) An array index can be of data type double. (1)
 - c) It's an error if an initializer list contains more initializers than there are array elements. (1)
 - d) Arrays can be passed as parameters to a function either by value or by reference. (1)
 - e) A function can return a value of type array. (1)
- 1.3. Explain what is wrong with the following code fragment and write code to correct it: (6)

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string prod_name[3]; // Assume product stock is populated...
    double price[3];

    // Find all products that cost more than R200
    for (int i = 0; i < 4; i++)
        cout << "Enter product name: ";
        cin >> prod_name;

    if (price[0] > 500.0)
        cout << prod_name[0] << " is expensive!\n";

    return 0;
}
```

Question 2 [20]

2.1. A lecturer at a college needs a simple system to manage and analyse student performance for a particular test. The lecturer wants to capture the scores of a group of students and quickly determine key statistics such as the highest score, lowest score, and average score.

Create a C++ program / system performs the following:

- Allow the lecturer to enter the number of students in the class (3)
- Store each student's score using a 1D array (3)
- Provide functionality to:
 - Input student scores (3)
 - Determine the maximum score (2)
 - Determine the minimum score (2)
 - Calculate the average score (2)

To make the system more flexible and user-friendly, the developer implements it using Object-Oriented Programming (OOP). A class is created to encapsulate all operations related to student scores.

Additionally, the system uses function overloading to display results in two ways:

- A basic view showing only the maximum score (2)
- A detailed report showing maximum, minimum, and average scores (3)

Question 3 [40]

3.1. There is an upcoming movie event at Carnival City, Johannesburg. Clients are required to reserve tickets through an online system. You have been requested to create a system that reads client's information from a file and display results to another file, respectively.

The information stored from a file include client's surname, name, movie title, movie type, seat number, movie price, and a ticket reference number. Create input file, namely **clients_info.dat** with the following data.

```
Venter, Wacko, Never Mind, Action, A_01, 180, 134QY78
Mthemu, Thando, Toughest Journey, Action, V_01, 220, 1267JJ67
Sally, White, The Revenge, Action, A_02, 180, 7877PX09
Pillay, Rohan, Avengers, Action, A_17, 150, 9037QS227
Black, Jason, Scream, Horror, H_01, 120, 9098AC881
Dlamini, Zinhle, Finding Nemo, Animation, K_05, 100, 9054EG445
Zulu, Sibusiso, Frozen, Animation, K_08, 102, 9021HJ118
Mabena, Thandi, Batman, Action, A_09, 145, 9010IK009
Motsepe, Nomsa, Dangerous Space, Horror, H_11, 135, 9091KM881
```

```

Sithole, Zanele, Frozen, Animation, K_20, 102, 9000TV990
Dube, Sibusiso, Batman, Action, A_21, 145, 9091UW881
Mkhabela, Kabelo, It Hurts, Horror, H_23, 135, 9073WY663
Zulu, Lerato, Coco, Animation, K_27, 105, 9037AC227
Ndlela, Nonhlanhla, Avengers, Action, A_29, 150, 9019CE009
Motsepe, Thabo, Finding Dorcus, Animation, K_30, 97, 9000DF990
Pillay, Zanele, Frozen, Animation, K_32, 102, 9082FH772
Radebe, Sibusiso, Batman, Action, A_33, 145, 9073GI663
Smith, Kabelo, It Hurts, Horror, H_35, 135, 9055IK445
Mthembu, Lerato, Coco, Animation, K_39, 105, 9019MO009
Mokoena, Themba, Joker, Action, A_40, 140, 9000NP990
Mabena, Thabo, Finding Dorcus, Animation, K_42, 97, 9082PR772
Ndlela, Bongani, Parasite, Animation, K_43, 132, 9073QS663
Motsepe, Zanele, Frozen, Animation, K_44, 102, 9064RT554
Radebe, Sibusiso, Batman, Action, A_45, 145, 9055SU445
Pillay, Kabelo, It Hurts, Horror, R_47, 135, 9037UW227

```

Your program should prompt the user to input file name and output file name. There are 3 types of movies available for clients to choose from.

The program should do the following:

- Read client's data line by line (i.e., character by character) from the input file. Validate the files. (3)
- Write a function to count number of lines in an input file. (3)
- Write a function to output the number of words in the file. You can consider a "word" to be any text that is surrounded by whitespace (for example, a space, comma, carriage return, newline) or borders the beginning or end of the file. (5)
- Write a function that allows a user to count the number of occurrences of a particular character in this file. For example, if the user enters the character 'e' you should count all the occurrences of the character 'e' (both uppercase and lowercase) in the file. Include a loop in your program which allows the user to search for characters multiple times if they wish (this should be a yes/no question). To enable users to search for characters multiple times, write a function which opens the file, counts for the number of occurrences of the character, closes the file and returns the occurrence count. Place the prompt for the user to enter a character to search for into another function. Finally include an assert check so that when the user is prompted whether they wish to continue the program to count the occurrence of another character, **if they enter an answer other than 'y' or 'n'** (for yes or no), the assert terminates the program. (5)
- Write a function to replace the first two (2) digits of a reference number. (5)
- Write a function to count the number of a specific movie type. (3)
- Calculate the total amount of each movie type. (3)

- Calculate the total amount of all movie reservations. (3)
- Write a function to display client's name, surname, movie type, and movie reference number and total amount paid for each movie type to a different file, namely **movie_report.txt**. Sort the records in a descending order based on the surname. (5)
- Write a menu-driven program, use do while loop, and a switch statement to invoke the functions and to solve this problem. [NB. Use classes and arrays to solve the problem] (5)

Question 4 [13]

4.1. Mark the following statements as true or false.

- a) A recursive solution of a problem reduces the problem into smaller versions of itself. (1)
- b) A recursive function always returns a value. (1)
- c) A function is called directly recursive if it calls itself. (1)
- d) In the general case, the solution to the problem is obtained directly. (1)

4.2 Consider the following function:

```
int recVal(int a, int b)
{
    if (a < b)
        return a + b;
    else
        return recVal(a - b, b + 1);
}
```

What is the output of the following statements?

- a) `cout << recVal(128, 15) << endl;` (1)
- b) `cout << recVal(-10, 8) << endl;` (1)
- c) `cout << recVal(148, 78) << endl;` (1)
- d) `cout << recVal(25, 25) << endl;` (1)

4.3 Consider the following recursive function:

```
1 int recFunc(int num)
2 {
3     if (num == 0)
4         return 0;
5     else if (num < 0)
6         return (-num);
7     else
8         return (num - recFunc(num - 5));
9 }
```

- a) Identify the base case. (1)
- b) Identify the general case. (1)
- c) If **recFunc(58)** is a valid call, what is its value? If not, explain why. (1)
- d) If **recFunc(-24)** is a valid call, what is its value? If not, explain why. (1)
- e) If **recFunc(0)** is a valid call, what is its value? If not, explain why. (1)